

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19 CS 1101]
I M. Tech I Semester (R19) Regular Examinations
DIGITAL DATA COMMUNICATIONS
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Explain the principle of QAM with the help of a neat diagram.	1	5	8
	b).	What is a constellation diagram? Explain the constellation diagram of 16QAM.	1	1	7
		OR			
2.	a).	Explain the OSI model for networking in brief. How does it differ from TCP/IP model?	9	5	8
	b).	What are different types of digital data transmission modes? Explain with examples.	4	2	7
		UNIT-II			
3.	a).	With the help of a neat diagram, explain the DTE-DCE interface.	2	5	8
	b).	Explain the various network topologies and compare the performance of each	3	5	7
		OR			
4.	a).	What are the various types of error correction codes? Explain any one.	6	2	8
	b).	Compare the synchronous and asynchronous data link protocols.	6	5	7
		UNIT-III			
5.	a).	What are the different types of packet switching? Explain.	8	2	8
	b).	Explain the architecture of IEEE 802.6.	9	2	7
		OR			
6.	a).	What are the different functions of bridges in the network? Explain various types of bridges used in the network.	7	1,5	8
	b).	Compare the time division and frequency division multiplexing techniques.	7	5	7
		UNIT-IV			
7.	a).	What is the need for multiplexing and explain various types of multiplexing techniques?	1	2	8
	b).	Explain various LAN topologies.	1	5	7
		OR			
8.	a).	Compare various switching techniques in detail.	8	5	8
	b).	Explain the function of various internetworking devices.	8	5	7
		UNIT-V			
9.	a).	Briefly explain various multiple access techniques.	1	5	8
	b).	What are the demerits in OFDMA? Explain how they can be overcome?	10	2	7
		OR			
10.	a).	Explain the principle and working advantages and applications of	10	5	8

		CDMA.			
	b).	Explain the principle and working of CSMA/CA technique in detail.	7	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS1102]

IM. Tech I Semester (R19) Regular Examinations
ADVANCED DIGITAL SIGNAL PROCESSING
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Using Decimation in Time algorithm find the FFT of $x[n]=\{1,2,3,1,-1,3,-4\}$.	1	2	8
	b).	Perform the two-band poly-phase decomposition of the transfer function $H(z) = \frac{2 + z^{-1}}{1 + 0.5z^{-1}}$	1	1	7
		OR			
2.	a).	Describe the frequency domain description of an Interpolator.	1	1	8
	b).	Calculate the DFT of $x[n]=\{1,0,1,2,1,3,1,4\}$.	1	2	7
		UNIT-II			
3.	a).	Describe the oversampling A/D and D/A converters.	2	3	8
	b).	Explain the implementation of sampling rate converters using Polyphase filters.	2	3	7
		OR			
4.	a).	Explain how sampling rate conversion of band pass signals can be achieved.	2	4	8
	b).	Draw and explain the block diagram of two channel QMF bank.	2	3	7
		UNIT-III			
5.	a).	Explain the blackmann and tukey method of smoothing the periodogram.	3	4	8
	b).	Derive the expressions for quality factor for various non-parametric methods.	3	3	7
		OR			
6.	a).	Discuss in brief about Bartlett method of power spectrum estimation.	3	6	8
	b).	Explain the Welch's method for the power spectrum estimation.	3	2	7
		UNIT-IV			
7.	a).	Explain the operation of Backward Linear predictor with a neat derivation.	4	3	8
	b).	what are the advantages of lattice structures?	4	2	7
		OR			
8.	a).	Explain lattice structures for IIR filters.	4	6	8
	b).	Explain the quantization and round off errors in a digital filter.	4	5	7
		UNIT-V			
9.	a).	Explain Burg method of power spectrum estimation.	5	4	8
	b).	Explain the autocorrelation function with its properties.	5	5	7
		OR			
10.	a).	Explain the MA & ARMA models for power spectrum estimation.	5	3	7
	b).	Describe the Yule-Walker method of power spectrum estimation.	5	4	8

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

[M19CS1103]

I M. Tech I Semester (R19) Regular Examinations

RADAR SIGNAL PROCESSING

DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Draw the block diagram of a Radar and explain	1	1	8
	b).	Obtain the frequency response of a Matched filter with non-white noise	1	2	7
		OR			
2.	a).	Discuss the efficiency of a non-matched filter	1	2	8
	b).	Explain the role of beacon and repeater expressions n Radar.	1	2	7
		UNIT-II			
3.	a).	Explain the I,Q detector with a block diagram	2	2	8
	b).	What is CFAR and explain average CFAR technique.	2	1	7
		OR			
4.	a).	Explain Likelihood-Ratio Receiver, Inverse Probability Receiver, and Sequential Observer,	2	2	8
	b).	Discuss Radar Signal Management–Schematics	2	2	7
		UNIT-III			
5.	a).	Obtain the ambiguity diagram for a pulse train consisting of five pulses	3	3	8
	b).	Explain the optimization for detecting signals in clutter when the relative Doppler shift is zero or unknown	3	4	7
		OR			
6.	a).	Discuss Waveform Design Requirements	3	3	8
	b).	Write short note on Family of Radar Waveforms.	3	2	7
		UNIT-IV			
7.	a).	What are the advantages of the thumb lack ambiguity diagram? And sketch the ideal thumb lack ambiguity diagram by a noise like waveforms.	4	1	8
	b).	Explain the basic principle of the linear FM pulse compression	4	3	7
		OR			
8.	a).	Discuss the applications, advantages and disadvantages of short pulse in a radar	4	3	8
	b).	Explain the SAW pulse compression in detail	4	4	7
		UNIT-V			
9.	a).	Discuss the principle of the binary phase coded pulse compression	5	5	8
	b).	Explain the properties of the frank poly phase codes.	5	5	7
		OR			
10.	a).	Write short note on Barker code	5	5	8
	b).	Write short note Maximal length sequences using PN codes	5	5	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS1104]

I M. Tech I Semester (R19) Regular Examinations

RF CIRCUIT DESIGN

ELECTRONICS AND COMMUNICATION ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Explain why skin effects are considered important for RF circuit design.	1	1	8
	b).	Describe briefly how passive components are realized on printed circuit boards at RF.	1	2	7
		OR			
2.	a).	Discuss Inductors and Capacitors - Voltage and Current in RF circuits	1	2	8
	b).	Write short note on Tuned RF / IF Transformers.	1	2	7
		UNIT-II			
3.	a).	Compare the characteristics of coaxial line, two-wire line and a parallel plate transmission line.	2	2	8
	b).	An input impedance of 25Ω of $\lambda / 4$ transformer is to be matched to a 50Ω micro strip transmission line at 500 MHz compute the length, width and characteristic impedance of the quarter-wave parallel plate transmission line. The thickness and relative dielectric constant of the substrate material are given as 1 mm and 4.0 respectively. Make assumptions if necessary.	2	1	7
		OR			
4.	a).	Define the following terms for a transmission line. i) Standing wave ratio ii) Return loss iii) Power in dBm iv) characteristic impedance.	2	2	8
	b).	Derive the expression for characteristic impedance of a short circuited transmission line.	2	2	7
		UNIT-III			
5.	a).	Describe how a tunable RF active filter can be realized.	3	3	8
	b).	With the help of neat diagrams describe the structure and functioning of a HEMT.	3	4	7
		OR			
6.	a).	Write a short note on low noise, linear RF BJT operation based on its structure?	3	3	8
	b).	Enumerate the importance of 'power relations' in the design of an amplifier at high frequencies.	3	2	7
		UNIT-IV			
7.	a).	Explain how stable performance can be assured for an RF transistor amplifier using corresponding stability circles.	4	1	8
	b).	Draw a neat circuit diagrams to explain how gain-bandwidth product limitation can be overcome in an RF broadband amplifier design.	4	3	7
		OR			
8.	a).	Discuss MMIC amplifiers,	4	3	8

	b).	Explain Low noise amplifiers	4	4	7
		UNIT-V			
9.	a).	With the help of neat schematics explain how oscillators should be configured to obtain high frequencies of oscillation. State the principle of operation. State the principle of operation of a dielectric resonator oscillator.	5	5	8
	b).	Low phase noise oscillator design	5	5	7
		OR			
10.	a).	Image Reject and Harmonic mixers, Frequency domain considerations.	5	5	8
	b).	Discuss about VCOs and Crystal Oscillators	5	5	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19 CS 1105]

I M. Tech I Semester (R19) Regular Examinations

ADVANCED COMPUTER NETWORKS

Department: OF ELECTRONICS & COMMUNICATIONS ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	How are congestion control and quality of service related?	1	K1	6
	b).	What is the difference between open-loop congestion control and closed-loop congestion control?	1	K1	9
		OR			
2.	a).	Explain about X.25 layers and frame relay operation?	1	K2	8
	b).	Briefly discuss about ATM layers. ?	1	K6	7
		UNIT-II			
3.	a).	Describe the physical configuration of SONET with required diagrams?	2	K6	9
	b).	Explain folding algorithm in interconnection networks?	2	K5	6
		OR			
4.	a).	Describe the structure of feedback shift register for generating PN sequence?	2	K6	9
	b).	Briefly explain the significance of Virtual private networks?	4	K6	6
		UNIT-III			
5.	a).	Explain about MAC sub layer and its addressing mechanism?	4	K5	8
	b).	Write short notes on frequency reuse and channel assignment strategies of cellular networks?	2	K2	7
		OR			
6.	a).	Explain the four general techniques to improve quality of service?	3	K2	9
	b).	Compare IPv4 and IPv6?	2	K2	6
		UNIT-IV			
7.	a).	Explain DSSS system with neat block diagram?	3	K2	8
	b).	Explain about Interconnection networks and their properties?	5	K2	7
		OR			
8.		With neat sketches explain about Bluetooth architecture and its layers?	5	K2	14
		UNIT-V			
9.	a).	Consider a wired point-to-point network. List some performance measures.	4	K1	8
	b).	Explain about GSM and GPRS	4	K2	7
		OR			
10.	a).	Explain the principle, advantages and applications of CDMA.	3		8
	b).	What are the demerits of CDMA? Explain.	3		7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS1106]
II M. Tech I Semester (R19) Regular Examinations
WIRELESS LANs AND PANs
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Discuss about the technical issues in wireless communications	CO1	K6	8
	b).	Explain about pure ALOHA in detail	CO1	K1	7
		OR			
2.	a).	Write about slotted ALOHA.	CO1	K2	8
	b).	Explain in detail about Wireless 4G systems.	CO1	K1	7
		UNIT-II			
3.	a).	Explain the importance of Wireless LANs.	CO2	K1	8
	b).	Explain in detail about Infrared technology.	CO2	K2	7
		OR			
4.	a).	Compare difference between wired and Wireless LAN.	CO2	K4	8
	b).	Write about Spread Spectrum technologies.	CO2	K1	7
		UNIT-III			
5.	a).	Explain in detail about different MAC layer issues.	CO3	K1	8
	b).	Explain about medium access control layer in wireless LAN.	CO3	K1	7
		OR			
6.	a).	Explain about different layers in IEEE 802.1 wireless LAN?	CO3	K1	8
	b).	How congestion can be controlled and avoided in MAC layer?	CO3	K1	7
		UNIT-IV			
7.	a).	Explain about different Blue Tooth Technologies.	CO4	K1	8
	b).	Write about different interference issues in Blue Tooth.	CO4	K1	7
		OR			
8.	a).	Compare Intra and Inter Pico net scheduling.	CO4	K4	8
	b).	Discuss different security issues in Blue tooth	CO4	K6	7
		UNIT-V			
9.	a).	Explain in brief about different Zig Bee technologies	CO4	K1	8
	b).	Write about IEEE 802.15.4 LR-WPAN Device architecture	CO4	K1	7
		OR			
10.	a).	Explain about IEEE 802.15.3a Ultra wideband?	CO5	K2	8
	b).	Write about different IEEE 802.15 working Group for WPAN.	CO5	K1	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS1107]

I M. Tech I Semester (R19) Regular Examinations
MOBILE COMPUTING TECHNOLOGIES
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Discuss the design considerations for mobile computing?	CO1	2	8
	b).	Explain Mobile Computing through Internet.	CO1	2	7
		OR			
2.	a).	Explain the three tier architecture of Mobile computing with neat diagram.	CO1	2	8
	b).	Explain Middle ware and Gateways in details.	CO1	2	7
		UNIT-II			
3.	a).	Discuss about the mobile services and data services in GSM.	CO2	2	8
	b).	Describe in detail about GPRS Network Architecture.	CO2	4	7
		OR			
4.	a).	Write in detail about Code division multiple accesses (CDMA).	CO2	4	7
	b).	What are the benefits of reservation schemes? How are collisions avoided during data transmission, why is the probability of collisions lower compared to classical Aloha? What are the disadvantages of reservation schemes?	CO2	1,2	8
		UNIT-III			
5.	a).	Explain the wireless LAN advantages?	CO3	4	8
	b).	Write a short note on IEEE 802.11 Standards.	CO3	2	7
		OR			
6.	a).	Write about Wireless LAN Architecture with neat sketch.	CO3	2	8
	b).	Discuss about SS 7 Signaling system?	CO3	1	7
		UNIT-IV			
7.	a).	Explain in detail about Security considerations in MIDP?	CO4	2	8
	b).	Write about Optional Packages?	CO4	2	7
		OR			
8.	a).	Draw Symbian OS Architecture and explain in detail.	CO4	2	8
	b).	Discuss in detail about Design Constraints in Applications for Handheld Devices.	CO4	2	7
		UNIT-V			
9.	a).	Compare H.323 and SIP?	CO5	2	8
	b).	Write any two applications of Voice over IP.	CO5	2	7
		OR			
10.	a).	What are different Security Techniques and Algorithms? Explain any one algorithm.	CO6	1	8
	b).	What are Security frameworks for Mobile Environment? Discuss.	CO6	1	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS1108]
I M. Tech I Semester (R19) Regular Examinations
NETWORK SECURITY AND CRYPTOGRAPHY
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Describe the relationship between security services and mechanisms	CO1	2	8
	b).	What are the different types of attacks? Explain with examples.	CO1	1	7
		OR			
2.	a).	What is double DES? What kind of attack is possible on double DES?	CO1	1	8
	b).	Explain about Block Cipher Principles?	CO1	2	7
		UNIT-II			
3.	a).	Explain digital signature standard algorithm for authentication?	CO2	2	8
	b).	Explain about International Data Encryption algorithm?	CO2	4	7
		OR			
4.	a).	Explain in detail about the Characteristics of Advanced Symmetric block ciphers.	CO2	2	8
	b).	Write short note on Traffic confidentiality, Key distribution.	CO2	2	7
		UNIT-III			
5.	a).	Perform encryption/decryption using RSA algorithm for the following: $p = 3, q = 11, e = 7, m = 5$	CO3	6	8
	b).	What are the main differences between Diffie-Hellman Key exchange, Elliptic Curve Cryptography.	CO3	2	7
		OR			
6.	a).	Explain about Discrete logarithms.	CO3	2	8
	b).	State Chinese remainder theorem and explain with one example.	CO3	1	7
		UNIT-IV			
7.	a).	Describe the different cryptographic functions provided by PGP.	CO4	2	8
	b).	What is MIME? What are the different content types of MIME?	CO4	1	7
		OR			
8.	a).	Explain digital signature standard algorithm for authentication?	CO4	2	8
	b).	What are HASH functions and how they are helpful in Message Authentications?	CO4	1	7
		UNIT-V			
9.	a).	Explain in detail the architecture of IP Security.	CO4	2	8
	b).	How web security can be achieved? What are the different mechanisms?	CO4	2	7
		OR			
10.	a).	Define the three classes of intruders and mention the intrusion techniques to protect from the intruders.	CO4	1	8
	b).	Explain the different types of viruses.	CO4	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19RD1101]

I M. Tech I Semester (R19) Regular Examinations
RESEARCH METHODOLOGY AND IPR
Common to CST,CS,PSA,IT & CAD
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Write briefly about good Research criteria.	CO1	K2	8
	b).	What are the errors in selecting a research problem?	CO1	K2	7
		OR			
2.		Describe briefly the Research process with neat sketch.	CO1	K2	15
		UNIT-II			
3.	a).	Write briefly about Effective Literature studies approaches.	CO1	K2	8
	b).	Explain about Research ethics.	CO3	K2	7
		OR			
4.	a).	Write briefly about Effective technical writing.	CO2	K3	8
	b).	Explain about the Format of research proposal.	CO2	K3	7
		UNIT-III			
5.		Write about the various steps in acquisition of trademarks rights.	CO5	K2	15
		OR			
6.	a).	Write briefly about International cooperation on Intellectual Property.	CO5	K2	8
	b).	Explain the procedure for grants of patents.	CO5	K2	7
		UNIT-IV			
7.		Explain about patent information and databases.	CO5	K2	15
		OR			
8.	a).	Write briefly about scope of patent rights.	CO5	K2	8
	b).	Write briefly about Licensing and transfer of technology.	CO4	K2	7
		UNIT-V			
9.		Write briefly about Administration in patent system.	CO5	K2	15
		OR			
10.		Write briefly about New developments in IPR.	CO6	K2	15

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19CS1201]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech II Semester (R19) Regular Examinations
WIRELESS COMMUNICATION AND NETWORKS
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Discuss in detail about Co-Channel and Adjacent Channel interference.	1	K2	8
	b).	What is the need of frequency reuse? Explain about different methods used for frequency reuse concept.	1	K1	7
		OR			
2.	a).	What is meant by hand off? Describe the classification of hand off process.	1	K1	8
	b).	Explain about the trunk and grade service in wireless communications. Illustrate various methods for improving coverage and Capacity in cellular system.	1	K5	7
		UNIT-II			
3.	a).	Explain the free space propagation model and write the expression for the received power.	2	K2	8
	b).	Describe the concept of reflection, diffraction and scattering in detail.	2	K2	7
		OR			
4.	a).	Discuss about different types of indoor propagation models.	2	K2	8
	b).	Explain in detail various outdoor propagation models	2	K5	7
		UNIT-III			
5.	a).	What is fading? How its effect the wireless mobile communication explain	3	K1	8
	b).	What is small-scale fading? Write the factors influencing fading.	3	K1	7
		OR			
6.	a).	Briefly explain fading due to multipath and Doppler spread..	3	K5	8
	b).	Briefly explain various Statistical Models for multipath Fading Channels.	3	K5	7
		UNIT-IV			
7.	a).	Explain about Equalization in a communication Receiver and explain about various Equalization techniques.	4	K5	8
	b).	Write about various diversity methods	4	K1	7
		OR			
8.	a).	Briefly explain Least Mean Square (LMS) and Recursive Least square (RLS) algorithms	4	K5	8
	b).	Explain Maximum Likelihood Sequence Estimation (MLSE) Equalizer	4	K2	7

		UNIT-V			
9.	a).	List out various advantages and Disadvantages of wireless LAN's	5	K2	8
	b).	Explain various topologies in LAN's	5	K5	7
		OR			
10.	a).	Write about IEEE802.11 Media Access Control standards and compare various IEEE 802.11 standards	55	K2	8
	b).	Write about IEEE802.16, Wireless PAN's and Hyper LAN	5	K2	7
CO-COURSE OUTCOME		KL-KNOWLEDGE LEVEL	M-MARKS		

[M19CS1202]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
II M. Tech II Semester (R19) Regular Examinations
IMAGE AND VIDEO PROCESSING
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M																		
		UNIT-I																					
1.	a).	Explain the process of image sampling and quantization?	CO1	K2	8																		
	b).	Discuss how image is analyzed using 2D Fourier transform?	CO2	K6	7																		
		OR																					
2.	a).	Discuss about KL transform and list out its merits and demerits?	CO1	K6	8																		
	b).	Interpret the basic relationships between pixels?	CO1	K2	7																		
		UNIT-II																					
3.	a).	Discuss about various frequency domain techniques of image enhancement?	CO3	K6	8																		
	b).	Explain the Weiner Filtering method?	CO1	K2	7																		
		OR																					
4.	a).	Elaborate different adaptive filters used for image restoration?	CO1	K6	8																		
	b).	Perform histogram equalization of an image whose pixel intensity distribution is given as below, <table border="1"><tr><td>Gray level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of pixels</td><td>790</td><td>1024</td><td>900</td><td>606</td><td>329</td><td>247</td><td>120</td><td>80</td></tr></table>	Gray level	0	1	2	3	4	5	6	7	No. of pixels	790	1024	900	606	329	247	120	80	CO4	K5	7
Gray level	0	1	2	3	4	5	6	7															
No. of pixels	790	1024	900	606	329	247	120	80															
		UNIT-III																					
5.	a).	Explain region based segmentation process?	CO2	K5	8																		
	b).	Discuss about lossy and lossless predictive coding models?	CO2	K6	7																		
		OR																					
6.	a).	Explain the process of edge detection while performing image segmentation	CO5	K2	8																		
	b).	Construct the Huffman code for the word “BABY”, and also compute the efficiency of Huffman code.	CO3	K3	7																		
		UNIT-IV																					
7.	a).	Discuss about photometric image formation model?	CO4	K6	87																		
	b).	How sampling is done for video signals?	CO1	K1	7																		
		OR																					
8.	a).	Explain about geometric image formation model	CO4	K2	8																		
	b).	Compare analog video and digital video signals?	CO1	K4	7																		

		UNIT-V			
9.	a).	Explain the mesh based motion estimation model?	CO4	K2	8
	b).	Discuss about predictive coding of video processing?	CO5	K6	7
		OR			
10.	a).	What is optical flow? Explain general methodologies of motion estimation	CO4	K1	8
	b).	Explain about region based motion estimation model?	CO4	K2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19CS1203]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
SOFT COMPUTING TECHNIQUES
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	What are AI Approaches and Explain?	CO1	2	8 M
	b).	Explain Expert Systems.	CO1	2	7 M
		OR			
2.	a).	Differentiate Intelligent control system and classic control system.	CO1	4	8 M
	b).	Write a short note on Symbolic Reasoning Systems.	CO1	1	7 M
		UNIT-II			
3.	a).	What is McCulloch-Pitts Neuron model? Explain it with one example?	CO2	2	8 M
	b).	What are the main differences among McCulloch-Pitts neuron model, Perceptron and Adeline.	CO2	4	7 M
		OR			
4.	a).	Explain in detail about Feed-Forward Multilayer Perceptron.	CO2	2	8 M
	b).	What are the main differences between Fourier transformation and Wavelet transformations?	CO2	4	7 M
		UNIT-III			
5.	a).	Differentiate Crisp set and Fuzzy set operation.	CO3	4	8 M
	b).	Explain any one application of Fuzzy System.	CO3	2	7 M
		OR			
6.	a).	Explain about Defuzzification techniques?	CO3	2	8M
	b).	Describe in detail about Fuzzy set operation?	CO3	2	7 M
		UNIT-IV			
7.	a).	Explain the concept of Genetic Algorithm with the help of flowchart?	CO4	5	8 M
	b).	Explain basic concept of Tabu search technique for problem solving optimization problems.	CO4	3	7 M
		OR			
8.	a).	Describe the basic concept of ANT – colony search technique for problem solving optimization problems.	CO4	3	8 M
	b).	Write short note on Composite Laminates?	CO4	6	7 M
		UNIT-V			
9.	a).	Explain Fuzzy Logic as a Soft computing Tool?	CO5	2	8 M
	b).	Explain about Fuzzy logic controller?	CO5	2	7 M
		OR			
10.	a).	How the GA is used in Fuzzy logic controller design?	CO5	1	8 M
	b).	Explain Genetic Algorithm as a Soft computing Tool?	CO5	2	7 M

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19CS1204]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech II Semester (R19) Regular Examinations
INTERNET PROTOCOLS
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Discuss about the Delivery and Routing of IP Packets.	1	K2	8
	b).	Discuss about IP addressing, Sub netting and Super netting.	1	K2	7
		OR			
2.	a).	Differentiate between Point-to-point WAN and Switched WAN.	1	K3	8
	b).	Explain briefly the design of ARP.	1	K2	7
		UNIT-II			
3.	a).	Explain State Transition Diagram.	2	K2	8
	b).	Explain TCP design.	2	K2	7
		OR			
4.	a).	Discuss services and features of SCTP	3	K2	8
	b).	Differentiate Indirect TCP, Snooping TCP and Mobile TCP.	3	K3	7
		UNIT-III			
5.	a).	Define Multicasting and explain Multicast routing with its applications.	3	K2	8
	b).	Explain the Hello message format of OSPF?	3	K2	7
		OR			
6.	a).	Explain Distance Vector routing and Link state routing.	4	K2	8
	b).	Explain the concept of Routing protocols	4	K2	7
		UNIT-IV			
7.	a).	Explain about HTTP.	3	K2	8
	b).	Write about DNS in the Internet.	3	K2	7
		OR			
8.	a).	Write about Communication in FTP.	3		8
	b).	Explain SNMP.	3	K2	7
				K2	
		UNIT-V			
9.	a).	Explain security in the Internet Firewalls.	4	K2	8
	b).	Explain Audio and Video Compression techniques.	4	K2	7
		OR			
10.	a).	Differentiate RTP, RTCP and Voice over IP	4	K3	8
	b).	Explain real time interactive audio/video processing	4	K2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19CS1205]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech II Semester (R19) Regular Examinations
CYBER SECURITY
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Sketch a matrix that shows the relationship between security mechanisms and attacks.	CO2	K3	8
	b).	What are the different RFCs and Internet standards related to security? Logically organize them and explain their contribution.	CO2	K5	7
		OR			
2.	a).	What is TCP session hijacking?	CO1	K2	8
	b).	Give a brief note on man-in-the-middle attacks.	CO3	K2	7
		UNIT-II			
3.	a).	What are the HMAC design objectives as per the RFC's? Explain.	CO4	K2	8
	b).	Explain any two approaches of Message Authentication.	CO1	K2	7
		OR			
4.	a).	Discuss the Secure Hash Function Algorithm in detail.	CO2	K2	8
	b).	Why do some block cipher modes of operation only use encryption while others use both encryption and decryption? Explain.	CO2	K4	7
		UNIT-III			
5.	a).	Use Fermat's Theorem to find a number x between 0 and 28 with x^{85} congruent to 6 modulo 29.	CO4	K3	8
	b).	What is digital signature? Explain the benefits of digital signature.	CO2	K2	7
		OR			
6.	a).	Explain the X.509 V3 certificate format	CO4	K2	8
	b).	How keys are generated in various cryptographic algorithms? Explain.	CO5	K2	7
		UNIT-IV			
7.	a).	In PGP, can an e-mail message use two different public key algorithms for encryption and signing? How is this defined in a message sent from Alice to Bob?	CO5	K2	8
	b).	Discuss how SSL record protocol provides confidentiality and message integrity for SSL connections?	CO6	K3	7
		OR			
8.	a).	With the help of neat diagram, explain the IP security architecture	CO1	K2	8
	b).	Explain Secure Electronic Transaction.	CO2	K2	7
		UNIT-V			
9.	a).	Consider any commercial hardware firewall and explain it in detail.	CO5	K3	8
	b).	What are the different intrusion detection techniques? How to prevent	CO5	K2	7

		false alarms in IDS?			
		OR			
10.	a).	Compare the various generations of Firewalls. Comment on the security achieved	CO5	K3	8
	b).	Select any three antivirus of your choice and explain their features.	CO6	K2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19CS1206]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech II Semester (R19) Regular Examinations
OPTICAL NETWORKS
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Discuss the SONET frame structures	1	K2	8
	b).	Explain the following functions of ATM i. Connections and cell forwarding ii. Signalling and Routing	1	K1	7
		OR			
2.	a).	Discuss different OADM architectures with neat figures.	1	K2	8
	b).	Draw the block diagram of broadcast and select ROADM and explain its functioning	1	K1	7
		UNIT-II			
3.	a).	Discuss cost trade-offs by considering PWDM ring architecture.	2	K2	8
	b).	Explain light path topology design (LTD) problem in brief.	2	K1	7
		OR			
4.	a).	Mention the key attributes of optical layer (light path) service.	2	K1	8
	b).	Explain adaptation management in WDM optical networks with neat diagrams	2	K2	7
		UNIT-III			
5.	a).	Explain various functions of network management.	3	K2	8
	b).	Illustrate services and interfacing in optical layers.	3	K2	7
		OR			
6.	a).	Explain multivendor interoperability.	3	K2	8
	b).	Explain performance and faulty management in optical networks.	3	K2	7
		UNIT-IV			
7.	a).	Explain unidirectional path switched rings used in the protection of SONET / SDH	4	K2	8
	b).	Discuss the protection in IP with an example.	4	K2	7
		OR			
8.	a).	Draw the block diagram of wavelength routing PON (WRPON) and explain its working	4	K2	8
	b).	Explain the functions of each element of an access network with architectural diagram	4	K2	7
		UNIT-V			
9.	a).	Explain optical time division multiplexing (OTDM) with a neat figure.	5	K2	8
	b).	Explain the following: i. Recirculation buffering ii. Burst Switching	5	K2	7
		OR			

10.	a).	Discuss architectural choices for next generation transport networks with neat diagrams.	5	K2	8
	b).	Explain how the transmission capacity of an optical link is increased by SDM	5	K2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

[M19 CS 1207]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
DIGITAL SIGNAL PROCESSORS AND ARCHITECTURE
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Explain the basic Digital signal processing system with the help of block diagram.	1	1	8
	b).	Explain in detail about sources of errors in DSP implementations.	1	2	7
		OR			
2.	a).	Discuss about the DFT and FFT calculations in Digital signal processors.	1	2	8
	b).	Write about fixed point and floating point formats.	1	2	7
		UNIT-II			
3.	a).	Discuss in brief about the data addressing capabilities of programmable DSP devices with examples.	2	1	8
	b).	Explain the operation of address generation unit in a DSP processor with neat block diagram.	2	3	7
		OR			
4.	a).	What are the features for external interfacing in a DSP processor?	2	4	8
	b).	Write about the programmability and program execution sequence of a DSP	2	4	7
		UNIT-III			
5.	a).	What is the need an interrupt in a processor? Write about Interrupts of TMS320C54XX Processor.	3	3	8
	b).	Explain the concept of Pipelining for speeding up the execution of an Instruction.	3	5	7
		OR			
6.	a).	Write about different on chip peripherals of TMS320C54XX Processor.	3	6	8
	b).	Explain about the memory space organization of TMS320C54XX Processor.	3	6	7
		UNIT-IV			
7.	a).	Explain the micro signal architecture in detail.	4	3	8
	b).	Discuss in brief about the basic peripherals in analog device family of DSP devices.	4	4	7
		OR			
8.	a).	Explain the architecture of Black fin processor with neat circuit diagram.	4	4	8
	b).	Explain the architecture of ADSP 2100.	4	4	7

		UNIT-V			
9.	a).	Draw and explain the block diagram of memory interface for TMS320C5416 processor.	5	3	8
	b).	How does DMA help in increasing the processing speed of a DSP processor?	5	2	7
		OR			
10.	a).	How an external bus can be interfaced with TMS320C54XX Processor? Explain.	5	5	8
	b).	Explain the Parallel I/O and Programmable I/O.	5	4	7
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

[M19CS1208]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech II Semester (R19) Regular Examinations
RADIO AND NAVIGATIONAL AIDS
Communication Systems
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Explain navigation equation and errors.	1	K2	8
	b).	Explain Hyperbolic navigation systems.	1	K2	7
		OR			
2.	a).	Explain TACAN and VORTAC.	1	K2	8
	b).	Describe Radio Navigation system types and performance parameters.	1	K2	7
		UNIT-II			
3.	a).	What is Inertial navigation System?	1	K2	8
	b).	Explain Gyro-cope in detail.	1	K2	7
		OR			
4.	a).	Explain Inertial Navigation mechanization.	2	K2	8
	b).	Explain Error analysis of Inertial Navigation system.	2	K2	7
		UNIT-III			
5.	a).	Explain geometric dilution of precision.	4	K2	8
	b).	Discuss satellite orbits and signal structure.	4	K2	7
		OR			
6.	a).	Differentiate Ionospheric group delay and tropospheric group delay.	1	K2	8
	b).	Explain Multipath errors.	1	K2	7
		UNIT-IV			
7.	a).	Explain wide area Augmentation system architecture	4	K2	8
	b).	Explain link budget and data capacity.	4	K2	7
		OR			
8.	a).	Explain precise positioning service local area DGPS	3	K2	8
	b).	Discuss about precision approach and error estimates	3	K2	7
		UNIT-V			
9.	a).	Write the applications of GPS	2	K2	8
	b).	Briefly explain Air and Land Navigation	3	K2	7
		OR			
10.	a).	Explain about geographical information systems.	4	K2	8
	b).	Explain about military and space navigation.	4	K2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19 CS 2101]
II M. Tech I Semester (R19) Regular Examinations
DETECTION AND ESTIMATION THEORY
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Define an Autoregressive Process. Calculate the values of mean, variance and autocorrelation values of a First order autoregressive model.	CO1	K1	8
	b).	Define shot noise. Calculate the values of mean, autocorrelation, and the power spectral density of the shot noise.	CO1	K1	7
		OR			
2.	a).	What is a random process? Write about Markov Processes and Gaussian processes.	CO1	K1	8
	b).	Explain any one of the procedure for the estimation of mean and variance.	CO1	K2	7
		UNIT-II			
3.	a).	What is Minimum Probability of Error? Explain Baye's decision rule.	CO2	K1	8
	b).	When does the LRT test under minimum probability of error criterion become identical to that under NP criterion?	CO2	K1	7
		OR			
4.	a).	Explain about the neyman-pearson classifier with necessary derivations.	CO2	K5	8
	b).	Discuss about Maximum Likelihood estimators.	CO2	K6	7
		UNIT-III			
5.	a).	Explain the operation of kalman filter with necessary block diagram	CO3	K2	8
	b).	What is a wiener filter and explain about one of its applications.	CO3	K2	7
		OR			
6.	a).	What is a minimum mean square error criterion? Differentiate between linear and nonlinear Minimum Mean Squared Error estimators.	CO3	K2	8
	b).	Explain about Innovations.	CO3	K2	7
		UNIT-IV			
7.	a).	Explain any one of the nonparametric estimator.	CO4	K2	8
	b).	What is the difference between point and interval estimators? Explain any one method of evaluating the interval estimators.	CO4	K1	7
		OR			
8.	a).	Explain the steps in the hypothesis testing of estimators.	CO4	K2	8
	b).	Explain briefly about Simple Linear Regression.	CO4	K2	7
		UNIT-V			
9.	a).	Explain the differences between Model-free Estimation and Model-	CO5	K5	8

		based Estimation of Autocorrelation Functions.			
	b).	Define Wide Sense Stationary and Strict Sense Stationary Random Processes. Explain the Tests for Stationary.	CO5	K1	7
		OR			
10.	a).	Derive the relationship between the autocorrelation and power spectral density.	CO5	K6	8
	b).	If an estimated ARMA model is, $X(N)=0.9X(n-1)-0.2X(n-2)+0.5e(n-1)+e(n)$. Find the corresponding autocorrelation function, autocorrelation coefficient, and power density spectrum.	CO5	K3	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19 CS 2102]
II/II M. Tech I Semester (R19) Regular Examinations
CODING THEORY AND APPLICATIONS
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER
TIME: 3Hrs. Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks.

			CO	KL	M
UNIT-I					
1.	a).	Write short notes on i) Logarithmic measure of information ii) Entropy	CO1	K1	8
	b).	The generator matrix of (6,3) block code is $G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$ then construct i) Parity check matrix 'H' ii) Standard array of this code	CO2	K3	7
OR					
2.	a).	Summarize the different types of errors and explain the control strategies?	CO1	K2	8
	b).	The parity check bits of a (8,4) block code are given by $P_1 = m_1 + m_2 + m_3$, $P_2 = m_0 + m_1 + m_2$, $P_3 = m_0 + m_1 + m_3$, $P_4 = m_0 + m_2 + m_3$, where m_1, m_2, m_3, m_4 are message bits. Find generator matrix and parity check matrix	CO4	K3	7
UNIT-II					
3.	a).	Design an encoder and encode the message vector $m=[1011]$ into a (7,4) code word using the generating polynomial $g(x) = 1+x^2+x^3$	CO5	K6	8
	b).	Explain about Cyclic Hamming codes	CO2	K2	7
OR					
4.	a).	List out and prove any five theorems related to cyclic codes.	CO1	K5	8
	b).	Explain encoding and decoding circuits of a systematic (n,k) linear codes.	CO3	K2	7
UNIT-III					
5.	a).	What are the applications of Convolutional codes in ARQ systems?	CO1	K1	8
	b).	Design encoder state diagram and identify the encoded output for $m=[10011]$ for a convolution code of rate 1/2, $K=3$, $g_0 = 1 \ 1 \ 1$, $g_1 = 1 \ 0 \ 1$	CO5	K6	7
OR					
6.	a).	Consider (2,1,2) code with $G(D)=[1+D^2, 1+D+D^2]$. Identify the GCD of its generator polynomial	CO3	K3	8
	b).	Examine the structural and distance properties of convolutional codes	CO2	K4	7
UNIT-IV					
7.	a).	Illustrate about Interleaved cyclic codes	CO1	K2	8
	b).	Explain single burst error correcting codes	CO3	K2	7
OR					
8.	a).	Elaborate decoding of single-burst error correcting cyclic codes	CO3	K6	8
	b).	Examine the bounds on burst error correcting capability of convolutional codes	CO1	K4	7
UNIT-V					
9.	a).	Discuss properties and characteristics for BCH codes	CO2	K6	8

	b).	Determine the generator polynomial for all primitive BCH codes of length 7.	CO4	K5	7
		OR			
10.	a).	Explain about error location polynomials for single and double error correction	CO4	K2	8
	b).	Construct syndrome computation circuit for double error correcting (15,7) BCH code	CO5	K6	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19CS2103]
M Tech III Semester (R19) Regular Examinations
SOFTWARE DEFINED RADIO
DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			CO	KL	M
		UNIT-I			
1.	a).	Explain the benefits and significance of SDR.	CO1	K1	8
	b).	Explain the Noise and Distortion in the RF chain.	CO1	K2	7
		OR			
2.	a).	Explain Digital filter banks.	CO1	K2	8
	b).	Discuss about Timing recovery in Digital Receivers using Multirate Digital filters.	CO1	K4	7
		UNIT-II			
3.	a).	Explain in detail about comparison of Direct Digital synthesis with Analog Signal synthesis.	CO2	K5	15
		OR			
4.	a).	Describe in detail about Generation of Random sequences.	CO2	K4	8
	b).	Explain parameters of ideal data converters.	CO2	K2	7
		UNIT-III			
5.	a).	Discuss in detail about ADC and DAC with neat architectures.	CO3	K6	15
		OR			
6.	a).	Explain about FPGAs.	CO3	K2	8
	b).	Write short note on Power management issues using a combination of DSPs, FPGAs, and ASICs.	CO3	K4	7
		UNIT-IV			
7.	a).	Explain about Object Oriented Programming.	CO4	K1	8
	b).	Write a short notes Joint Tactical Radio System.	CO4	K2	7
		OR			
8.	a).	Explain SDR-3000 Digital Transceiver Subsystem.	CO4	K4	8
	b).	Discuss about CHARIOT.	CO4	K6	7
		UNIT-V			
9.	a).	Write a short notes WITS	CO5	K1	8
	b).	Explain about object brokers	CO5	K2	7
		OR			
10.	a).	Discuss about speak easy systems	CO5	K6	8
	b).	Write a short notes SDR-3000 Digital Transceiver Subsystem	CO5	K1	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS